Independent effects of mental fatigue and drop height on drop jump performance in American football athletes: an exploration of central-peripheral interaction

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ABSTRACT

Background. This study aimed to investigate the effects of mental fatigue (MF) and different drop heights on the athletic performance of American football players executing the drop jump (DJ) movement.

Methods. Twelve male American football athletes were selected as subjects. The Vicon infrared three-dimensional motion capture and analysis system, Kistler three-dimensional force platform, and other instruments were used. MF was induced through a Stroop task, and the DJ performance of the subjects was tested at drop heights of 30 cm, 40 cm, and 50 cm. A 2 × 3 repeated measures analysis of variance was employed.

Results. Regarding different jump heights, both under MF and baseline conditions, 40 cm and 50 cm heights resulted in lower jump heights compared to the 30 cm height ($p = 0.002$, $p = 0.008$); in terms of the rate of force development (RFD) metric, both under MF and baseline conditions, 40 cm and 50 cm heights showed lower RFD compared to the 30 cm height ($p < 0.001$); in the average power output metric, compared to the baseline, MF resulted in lower average power output across different drop heights ($p < 0.001$); in the reactive strength index (RSI) metric, compared to the baseline, MF resulted in lower RSI across different drop heights ($p = 0.001$), and compared to the 30 cm height, 40 cm and 50 cm heights showed lower RSI ($p = 0.004$, $p = 0.008$); in the Reactive Strength Ratio (RSR) metric, compared to the baseline, MF resulted in lower RSR across different drop heights ($p = 0.001$); in the K_{leg} metric, compared to the 30 cm height, 40 cm and 50 cm heights showed higher K_{leg} ($p = 0.001$, $p = 0.008$).

Conclusions. Under MF conditions, athletes' performance in average power output, RSI, and RSR declined, suggesting a reduction in central nervous system efficiency. Additionally, increasing the drop height to 40 cm or above significantly reduced jump height and RFD, and increased K_{leg}. However, no interactive effects between MF and drop height were observed.

Subjects Kinesiology, Biomechanics, Sports Injury, Sports Medicine

Keywords Mental fatigue, Drop jump, Athletic performance, American football, Drop height

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INTRODUCTION

American football is a globally popular team sport, renowned for its high-intensity physical confrontations, complex tactical strategies, and the extreme demands it imposes on athletes' comprehensive physical fitness (*Hoffman, 2008*; *Liu et al., 2020*; *Robineau et al., 2017*). In competitive sports, the performance of American football players is influenced by numerous factors, including physical and psychological elements. This sport demands not only exceptional physical qualities from athletes but also rapid decision-making abilities and tactical comprehension. Consequently, mental fatigue (MF) may significantly impact athletes' on-field performance (*Mariano, Martin & Mara, 2023*). Particularly in the critical moments before a game, prolonged mental concentration and decision-making pressures can degrade cognitive functions, thereby impairing reaction speed, judgment accuracy, and tactical execution capabilities (*Schampheleer & Roelands, 2024*; *Staiano et al., 2024*).

MF typically refers to a reduction in the brain's information processing capacity following prolonged cognitive tasks, manifesting exhaustion and a lack of energy (*Marcora, Staiano & Manning, 2009*). Furthermore, as modern life accelerates and technological products become more prevalent, athletes encounter increased mental exertion outside of training and competition. For example, the frequent use of smartphones, electronic games, and sleep deprivation can all trigger MF, and recovering from this state of fatigue is challenging within a short timeframe (*Filipas et al., 2021*; *Thompson et al., 2019*; *Thompson et al., 2020*; *Smith et al., 2019*). Existing evidence demonstrates that MF can impair decision-making and technical skill execution in team invasion sports like soccer and basketball (*Gantois et al., 2020*; *Moreira et al., 2018*; *Smith et al., 2016*). Although these sports have different skill requirements compared to American football, they share notable similarities in match demands, such as high-intensity intermittent exercise and the need for sustained attention to the surrounding environment and teammates. These similarities imply that American football athletes might also be vulnerable to the negative effects of MF on performance. *Mariano, Martin & Mara (2023)* identified a significant negative correlation between MF and technical performance in a survey of 20 elite male rugby athletes, underscoring the importance of assessing MF in rugby athletes.

The drop jump (DJ) is a crucial test for evaluating lower limb explosive power and neuromuscular efficiency, with its execution precision directly reflecting an athlete's competitive state (*Leukel et al., 2008*; *Furuhashi et al., 2021*; *Yang et al., 2024*). Within sports science, the DJ test is widely utilized, as it effectively simulates the stretch-shortening cycle (SSC) in actual sports movements and efficiently assesses athletes' biomechanical characteristics and reactive strength in athletic performance (*Di Giminiani & Petricola, 2016*; *Matic et al., 2015*; *Horita et al., 2002*). *Mrdakovic et al. (2008)* and *Byrne et al. (2010)* investigated the effects of DJs at various drop heights (20 cm to 80 cm) on athletes' lower limb muscle activity and ground reaction forces, uncovering differences in biomechanical efficiency across drop heights and their substantial influence on athletic performance. Nevertheless, in practical training, DJs at drop heights of 30, 40, and 50 cm have become popular training tools for coaches and athletes due to their ease of use and safety. These heights serve as a practical method for assessing athletes' explosive power

and neuromuscular coordination (*Peng, 2011*; *Wang et al., 2021*). For American football athletes, DJ training at these drop heights enhances explosive power and agility while effectively mirroring the game's actual movement demands.

While extensive research exists on the effects of varying drop heights on athletes' DJ performance (*Mrdakovic et al., 2008*; *Byrne et al., 2010*; *Peng, 2011*; *Wang et al., 2021*; *Sotiropoulos et al., 2023*; *Ramirez-Campillo et al., 2022*), limited studies have examined how MF influences the performance of athletes in high-intensity, complex decision-making sports such as American football during DJ execution. Furthermore, the impact of MF on DJs at different drop heights remains underexplored. Previous research has demonstrated that MF can impair various aspects of physical performance. For instance, *Lima-Junior et al. (2024)* found that MF induced by smartphone use negatively affected inhibitory control, though it did not significantly impact countermovement jump performance in trained individuals. Similarly, *Queiros et al. (2021)* observed that MF reduced training volume in resistance exercises, indicating a potential influence on athletic performance. On the other hand, *Díaz-García et al. (2024)* highlighted that brain endurance training could mitigate the detrimental effects of MF on resistance exercise performance. Consequently, this study employs DJs at 30, 40, and 50 cm drop heights to investigate how MF affects the DJ performance of American Football athletes at these frequently used heights, offering novel insights into athletic performance under high-pressure conditions.

Given this context, this study aims to investigate how MF affects the athletic performance of American football athletes performing DJ movements at different drop heights. The study seeks to provide experimental data and theoretical insights regarding the effects of MF on DJ performance in American football athletes. Drawing on prior research, the following hypotheses are proposed: (1) MF has a lesser impact on low-height DJ than on high-height DJ; (2) MF does not significantly affect the jump height of American football athletes at different drop heights during DJ; (3) different drop heights significantly impact the execution of DJ movements by American football athletes.

MATERIALS & METHODS

Study design

This study employed a within-subjects repeated-measures design to investigate the independent effects of MF and different drop heights on the DJ performance of American football athletes. The design was chosen to specifically address the knowledge gap regarding how MF impacts the DJ performance of athletes in high-intensity, complex decision-making sports like American Football at frequently used drop heights. By incorporating an orientation phase for participant familiarization and a practice session to ensure understanding of experimental procedures, the study aimed to provide precise and reliable data. It included two main conditions: a MF intervention induced by a 45-minute Stroop task, and a control condition without MF intervention. DJ tests were conducted at varying drop heights both before and after the intervention to assess the influence of MF on athletic performance, thereby offering valuable insights for the training management, fatigue recovery, and competitive state optimization of American football athletes.

Participants

G*Power 3.1.9 software was utilized for sample size estimation. Based on prior studies (*Faul et al., 2007*; *Brysbaert, 2019*; *Griffin et al., 2021*) and the statistical methods of this research, parameters including power ($1-\beta$) of 0.80, Type I error rate (α) of 0.05, and effect size (f) of 0.4 were set. Consequently, a minimum required sample size of 10 participants was calculated. Ultimately, 12 male athletes from the Soochow University American football team were recruited. Their average age was 22.2 ± 1.9 years, height 181.4 ± 3.1 cm, body mass 80.1 ± 8.5 kg, and training experience 4.8 ± 1.5 years. All participants were key players in the China Collegiate American football League championship team, with extensive competitive experience at the national level. Their current training routine includes three to five sessions per week, encompassing technical drills, tactical practices, and physical conditioning, each lasting 2–3 h. Additionally, all subjects had no prior history of professional training in other sports, ensuring specialization in American football. They were thoroughly informed about the test procedures and potential risks, and provided written consent. The study was approved by the Ethics Committee of Soochow University (Approval No. SUDA20240626H03). The inclusion criteria were established to ensure the selection of a homogeneous sample with high physical fitness, free from lower limb joint or neuromuscular diseases or injuries in the past six months, similar morphological characteristics, no intense exercise 24 h before testing, no caffeine intake, abstinence from alcohol for at least one week before testing, and a stable psychological state without severe disorders or excessive stress. These criteria were designed to minimize variability and potential confounding factors, ensuring the sample's representativeness and suitability for the study.

Experimental equipment

Vicon infrared 3D motion capture and analysis system

The system consisted of eight infrared cameras (Model: MX13, Vicon Motion Systems, Oxford, UK), MX Net, MX Control, a PC workstation, calibration kits, and standard accessories. It operated at a sampling frequency of 100 Hz and used the Plug in Gait Full-Body model in Vicon Nexus software for motion tracking and data acquisition.

Kistler 3D force platform

Two Kistler 3D force platforms (90 cm × 60 cm × 10 cm, Model: 9281EA, Kistler Instruments, Winterthur, Switzerland) were embedded in the ground within the motion capture system's infrared camera range. They operated at a sampling frequency of 1000 Hz and were synchronized with the Vicon system *via* an analog-to-digital converter for data collection.

Study design and procedure

Prior to the commencement of each experimental session, participants underwent a familiarization phase that included practice trials of the DJ movement at all tested heights (30 cm, 40 cm, 50 cm) to ensure understanding and consistency. Participants were required to wear tight-fitting shorts and athletic shoes provided by the laboratory. All experiments were conducted in a controlled laboratory environment with temperature maintained

at 22–24 °C and relative humidity at 50–60%. Following the completion of warm-up activities, 27 markers with a diameter of 14 millimeters were affixed to the participants' body surfaces according to the full-body model protocol (Fig. 1). To ensure the accuracy of the experiment, the application of all markers was performed by the same experienced technician who is a PhD holder from Soochow University and a laboratory technician at the Sports Biomechanics Laboratory of Soochow University, with over 10 years of experience in motion capture studies. The equipment used included the Vicon infrared 3D motion capture system and Kistler 3D force platforms, as detailed in 'Experimental equipment'. Participants were instructed to execute each DJ with maximal effort, focusing on minimizing ground contact time and maximizing jump height. Data collection was synchronized between the motion capture and force platforms *via* an analog-to-digital converter, and raw data were processed using Visual 3D software with filtering parameters as described. All experimental sessions were conducted during the non-competitive season, specifically within the one-week transition phase following the competitive season. To minimize circadian rhythm effects, all sessions were scheduled between 15:00 and 19:00, as recommended by *Wang et al. (2025)*, to ensure task-induced fatigue predominated over physiological fluctuations.

Before the initiation of testing, a questionnaire was employed to assess the compliance of the participants. Subsequently, The National Aeronautics and Space Administration Task Load Index (NASA-TLX) Subjective Workload Scale was used to measure the participants' workload levels before the commencement of the test. After the completion of the MF task, the NASA-TLX and other indicators were measured again to assess the participants' workload levels (*Wang et al., 2025*). Following this, the participants performed an additional approximately 2-minute warm-up before immediately proceeding with the corresponding tests.

Participants stood on a plyometric box with hands on hips, awaiting instructions. Upon command, they stepped forward, fell freely, and performed a full DJ movement sequence: "landing-contact-jump-off-airborne-landing". The movement included contact and airborne phases, with the contact phase further divided into braking and push-off phases (Fig. 2). Athletes maintained balance and minimized ground contact time to maximize jump height. Each DJ test had a 30-second recovery interval, and three successful tests were collected for each drop height. The maximum jump height from the three trials was used for analysis.

Experimental protocol

The MF intervention involved participants engaging in a 45-minute Stroop task, designed based on the studies by *Lopes et al. (2020)* and *Badin et al. (2016)* and proven effective in inducing MF (*Mangin et al., 2022*). On a computer screen, the Chinese characters for "red", "green", "blue", and "yellow" appeared one at a time in random order, each displayed in one of the four colors. The probability of the character's color not matching its meaning was set at 50%, and participants pressed a button based on the character's color. Each character was displayed for 1,000 ms, followed by a 1,000 ms blank screen interval before the next character appeared. Participants completed a total of 1,350 judgment reactions

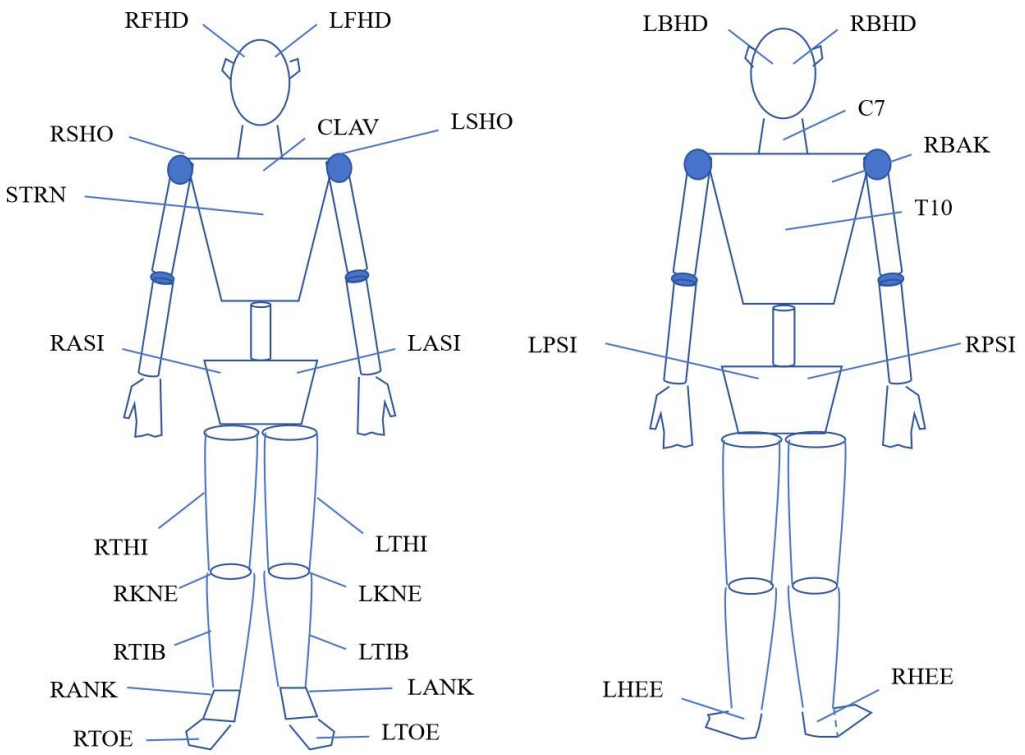

Figure 1 **Schematic diagram of marker attachment.** RFHD, Right Front Head (Temple); LFHD, Left Front Head (Temple); RSHO, Right Shoulder (Acromion); CLAV, Clavicle (Suprasternal Notch); LSHO, Left Shoulder (Acromion); STRN, Sternum (Xiphoid Process); RASI, Right Anterior Superior Iliac Spine; LASI, Left Anterior Superior Iliac Spine; RTHI, Right Thigh; LTHI, Left Thigh; RKNE, Right Knee (Lateral Epicondyle); LKNE, Left Knee (Lateral Epicondyle); RTIB, Right Tibia (Shank); LTIB, Left Tibia (Shank); RANK, Right Ankle (Lateral Malleolus); LANK, Left Ankle (Lateral Malleolus); RTOE, Right Toe (2nd Metatarsal Head); LTOE, Left Toe (2nd Metatarsal Head); LBHD, Left Back Head; RBHD, Right Back Head; RBAK, Right Back (Right Scapula); LPSI, Left Posterior Superior Iliac Spine; RPSI, Right Posterior Superior Iliac Spine; LHEE, Left Heel (Calcaneus); RHEE, Right Heel (Calcaneus).

Full-size DOI: 10.7717/peerj.20574/fig-1

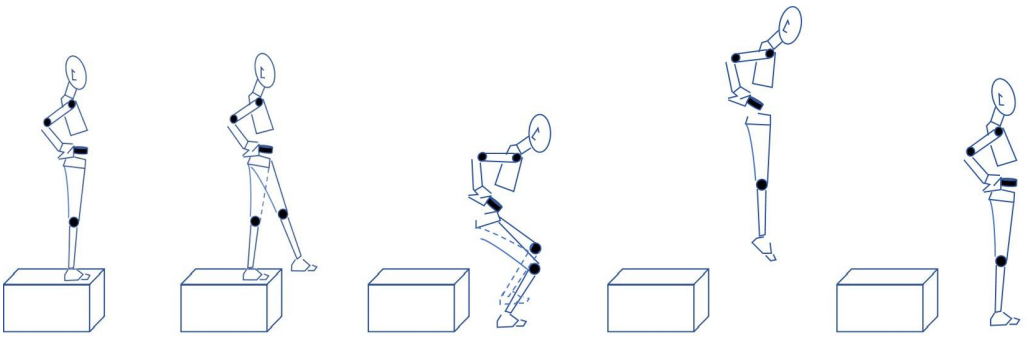

Figure 2 **Action diagram.**

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during the task. They were instructed to respond as quickly and accurately as possible. If a response was incorrect or no response was made within 1,500 ms, the system emitted a beep to encourage greater focus. The experiment took place in a separate, quiet room, with E-prime 3.0 software managing task operations. Two staff members supervised the process to ensure smooth progress and participant concentration. The parameters of the Stroop task, including character presentation time and the probability of color—meaning mismatch, were determined through reference to prior studies and pilot experiments. These parameters were optimized to impose sufficient cognitive load on participants, ensuring the effective induction of MF and providing a reliable foundation for subsequent assessments of its impact on athletic performance.

Data collection and analysis

Raw Vicon data were analyzed in Visual 3D software (C-Motion, Germantown, MD, USA) for kinematic and inverse dynamics analysis. A 4th-order low-pass Butterworth filter smoothed the three-dimensional coordinates (10 Hz) and force platform data (50 Hz). Takeoff and landing were defined as when the vertical ground reaction force (vGRF) was less than or exceeded 10 N (*Decker et al., 2003*; *Slater et al., 2015*; *Wang et al., 2024*).

The main observational indices in this study were:

(1) The maximum takeoff height normalized by body height (BH), calculated using the waist point method (*Wan et al., 2017*). This index reflects the athlete's vertical jumping ability during DJ, serving as a key measure of lower limb explosive power and athletic performance.

(2) The peak vGRF during the push-off phase ($vGRF_{peak}$, N). It represents the maximum vertical ground reaction force generated by the athlete during the push-off phase, reflecting the athlete's lower limb strength and ability to exert force on the ground

(3) The rate of force development (RFD, N/s) = $vGRF_{peak}/K_t$, where K_t is the time required to reach the peak force during the push-off phase (*Molina & Denadai, 2012*). RFD reflects the speed and efficiency with which muscles develop force, and is an important indicator of rapid strength and neuromuscular coordination. It helps assess the athlete's ability to exert force quickly during the push - off phase of DJ.

(4) Average power output (W/kg) = $(g^2 \times T_f \times T_t)/(4\times T_c)$; Reactive Strength Index (RSI, m/s) = H_f/T_c; Reactive Strength Ratio (RSR) = T_f/T_c, where g is the acceleration due to gravity (9.8 N/kg), T_f is the time in the air after the push-off phase ends, H_f is the takeoff height, T_c is the contact time from the first moment of foot contact to the moment of takeoff, and $T_t = T_f + T_c$ (*Bosco, Luhtanen & Komi, 1983*; *Healy, Kenny & Harrison, 2018*; *Healy, Kenny & Harrison, 2016*). These indices comprehensively evaluate the athlete's power output, reactive strength, and movement efficiency during DJ, providing a detailed assessment of athletic performance.

(5) Lower limb stiffness (K_{leg}, BW/m) = $Fz_{max}/\Delta L$, where Fz_{max} represents the maximum vertical ground reaction force normalized by body weight during the braking phase, and ΔL represents the maximum change in lower limb length, *i.e.,* the vertical displacement change of the hip joint center from the moment of contact to the completion of braking (*You & Huang, 2022*). K_{leg} reflects the stiffness characteristics of the lower limbs during

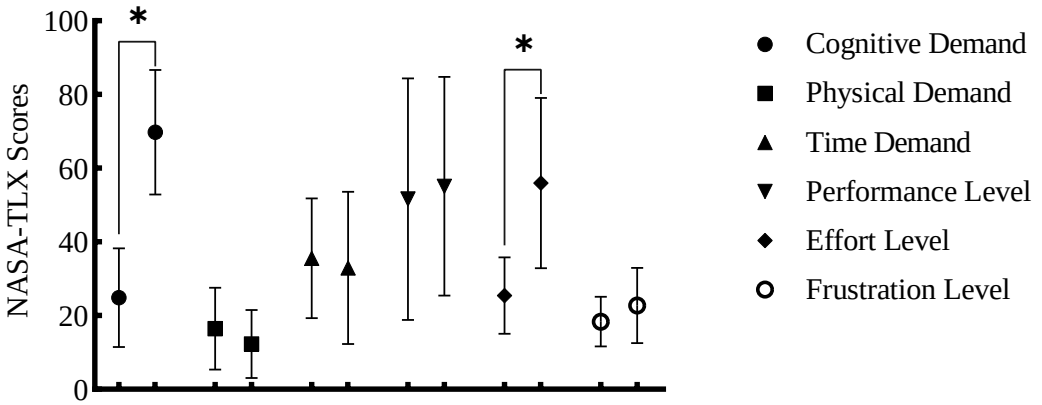

Figure 3 **NASA-TLX Scores before and after the MF.** *$p < 0.05$.
Full-size DOI: 10.7717/peerj.20574/fig-3

landing, which is related to the athlete's ability to absorb impact force and store elastic energy, and has a significant impact on DJ performance and injury prevention.

Statistical analysis

Data were analyzed using SPSS 26 (IBM Corp., Armonk, NY, USA). Descriptive statistics are reported as mean ± standard deviation (M ± SD). The Shapiro–Wilk test assessed normality, and Levene's test assessed homogeneity of variance. Paired-samples t-tests compared NASA-TLX scores pre- and post-intervention. A 2 × 3 (condition (baseline, MF) × height (30, 40, 50 cm)) repeated-measures ANOVA was conducted. Mauchly's test assessed sphericity; when violated, Greenhouse–Geisser corrections were applied. Statistical significance was set at $\alpha = 0.05$, with Bonferroni-adjusted *post hoc* tests.

RESULTS

MF intervention outcomes

In the MF protocol, there was a significant increase in the cognitive demand ($p < 0.001$) and effort expenditure ($p = 0.002$) of the participants, confirming the successful induction of MF (Fig. 3).

Effects of MF on jump height and $vGRF_{peak}$ during push-off phase

A significant main effect of drop height on jump height was observed ($p < 0.001$). *Post-hoc* comparisons revealed that, under both MF and baseline conditions, the heights of 40 cm and 50 cm resulted in lower jump heights compared to the 30 cm height ($p = 0.002$, $p = 0.008$). No statistically significant differences were observed for the $vGRF_{peak}$ measure ($p > 0.05$) (Fig. 4).

Effects of MF on RFD, average power output, RSI, RSR, and K_{leg}

A significant main effect of drop height on the RFD measure was observed ($p < 0.001$). *Post-hoc* comparisons revealed that, under both MF and baseline conditions, the 40 cm

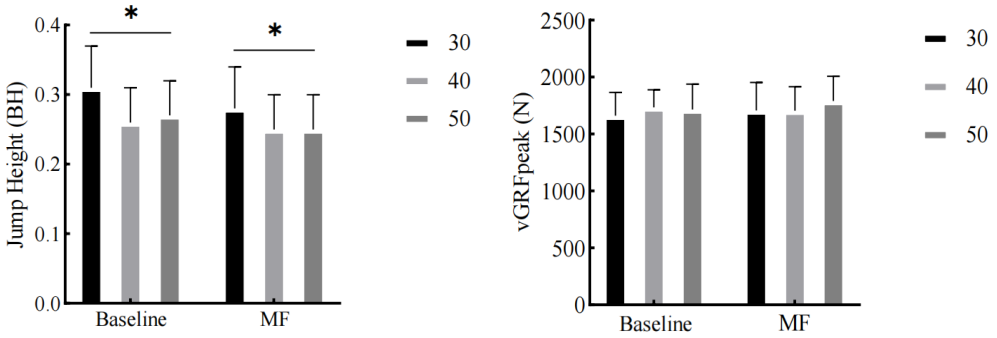

Figure 4 **The impact of MF on jump height and the vGRF$_{peak}$.** $^{*}p < 0{:}05$.
Full-size DOI: 10.7717/peerj.20574/fig-4

and 50 cm heights resulted in lower RFD compared to the 30 cm height ($p < 0.001$). A main effect of MF on average power output was observed, with lower average power output observed across different drop heights after MF compared to baseline ($p < 0.001$). For the RSI measure, main effects of MF and drop height were observed ($p = 0.001$ for both). *Post-hoc* comparisons indicated that, compared to baseline, lower RSI was observed after MF across different drop heights ($p = 0.001$), and lower RSI was observed at 40 cm and 50 cm heights compared to the 30 cm height ($p = 0.004$ and $p = 0.008$, respectively). A main effect of MF on RSR was observed, with lower RSR observed across different drop heights after MF compared to baseline ($p = 0.001$). A significant main effect of drop height on K_{leg} was observed ($p = 0.002$). *Post-hoc* comparisons revealed that, compared to the 30 cm height, higher K_{leg} was observed at 40 cm and 50 cm heights ($p = 0.001$ and $p = 0.008$, respectively) (Table 1).

DISCUSSION

The study results indicate that MF significantly affects the performance of American football athletes in executing DJs, particularly in terms of average power output, RSI and RSR, but not in jump height, vGRF$_{peak}$, RFD, or K_{leg}. Additionally, no interactive effects were found between MF and drop height, suggesting that MF does not differentially affect DJs at various drop heights tested in this study, contradicting our initial hypothesis. The impact of MF primarily lies in cognitive function and decision-making processes. Although MF does not directly lead to muscle fatigue, its negative effects on cognitive function and decision-making may indirectly affect athletic performance (*Smith et al., 2018*; *Smith et al., 2019*). Therefore, the decline in average power output, RSI, and RSR under MF may be associated with reduced efficiency of the central nervous system. This is consistent with the findings of *Brown et al. (2020)*, who concluded that MF impairs physical performance and thus affects movement execution. Furthermore, *Davidow et al. (2023)* found that MF significantly reduced tackling technique proficiency in amateur rugby union players, suggesting that impaired motor coordination under MF may stem from

Table 1 The impact of MF on average power output, RFD, RSI, RSR, and K^{leg}. $^{*}p < 0:05$.

Variables	Baseline	MF	Main effect		Interaction effect
			MF	Height	MF × Height
RFD (N/s)					
30 cm	9927.54 ± 1682.56	8590.23 ± 1433.50	$p = 0.057$	$\boldsymbol{p < 0.001^{*}}$	$p = 0.303$
40 cm	7246.60 ± 1752.35	6036.76 ± 1629.16	$F = 4.537$	$F = 26.070$	$F = 1.261$
50 cm	6885.17 ± 1767.35	6528.97 ± 1574.66	$Eta^2 = 0.292$	$Eta^2 = 0.703$	$Eta^2 = 0.103$
Average power output (W/kg)					
30 cm	29.94 ± 5.21	25.19 ± 3.86	$\boldsymbol{p < 0.001^{*}}$	$p = 0.189$	$p = 0.118$
40 cm	27.84 ± 5.25	24.78 ± 3.39	$F = 24.249$	$F = 1.797$	$F = 2.361$
50 cm	27.86 ± 5.79	24.45 ± 4.36	$Eta^2 = 0.688$	$Eta^2 = 0.140$	$Eta^2 = 0.177$
RSI (m/s)					
30 cm	1.10 ± 0.29	0.85 ± 0.19	$\boldsymbol{p = 0.001^{*}}$	$\boldsymbol{p = 0.001^{*}}$	$p = 0.322$
40 cm	0.88 ± 0.24	0.74 ± 0.18	$F = 20.462$	$F = 14.208$	$F = 1.131$
50 cm	0.87 ± 0.26	0.73 ± 0.18	$Eta^2 = 0.650$	$Eta^2 = 0.564$	$Eta^2 = 0.093$
RSR					
30 cm	1.07 ± 0.23	0.87 ± 0.16	$\boldsymbol{p = 0.001^{*}}$	$p = 0.511$	$p = 0.256$
40 cm	1.04 ± 0.21	0.89 ± 0.15	$F = 18.470$	$F = 0.692$	$F = 1.451$
50 cm	1.02 ± 0.23	0.87 ± 0.15	$Eta^2 = 0.627$	$Eta^2 = 0.059$	$Eta^2 = 0.117$
K_{leg} (BW/m)					
30 cm	40.02 ± 16.87	50.57 ± 8.96	$p = 0.499$	$\boldsymbol{p = 0.002^{*}}$	$p = 0.897$
40 cm	56.22 ± 13.84	61.72 ± 10.03	$F = 0.490$	$F = 11.782$	$F = 0.109$
50 cm	63.19 ± 19.99	66.26 ± 19.21	$Eta^2 = 0.043$	$Eta^2 = 0.517$	$Eta^2 = 0.010$

deficits in executive functions and suboptimal neural signaling. Therefore, in high-pressure environments, athletes' technical execution and decision-making abilities may be negatively affected by MF.

It is noteworthy that although MF had a certain negative impact on average power output, RSI, and RSR during the execution of DJs, it did not significantly affect jump height and RFD, which aligns with our hypothesis 2. Similar to previous findings by *Weerakkody et al. (2021)*, after a 30-minute Stroop test intervention on 25 amateur Australian football athletes, MF did not affect the jump height during either standing vertical jumps or running vertical jumps. The resistance of RFD to MF might be attributed to its dependence on intrinsic muscle–tendon unit properties and well-learned movement patterns, which could be less susceptible to cognitive fatigue compared to more complex, power-dependent outputs requiring optimal central drive and coordination. Considering that American football is a sport with high cognitive function demands, athletes need to maintain a high level of alertness and concentration during matches and training to capture and process information from the rapidly changing field environment (*MacDonald & Minahan, 2016*). This sustained cognitive load often induces MF, and frequent competitions and training further increase the frequency with which athletes are in a state of MF (*Abbott et al., 2020*). This frequent exposure necessitates that athletes develop effective psychological and behavioral coping strategies to manage MF. Research indicates that experienced athletes

often demonstrate superior abilities in regulating cognitive resources and maintaining performance under pressure (*Russell et al., 2020*; *Martin et al., 2016*). This suggests that extensive competition experience may make athletes more efficient in managing cognitive resources and coping with psychological stress, thereby maintaining better athletic performance under high-pressure competitive conditions. Therefore, the results of this study indicate that athletes may effectively manage stress and fatigue by developing psychological resilience and adopting effective coping strategies, thus maintaining their athletic performance to a certain extent in high-pressure environments. This finding emphasizes the importance of psychological skill training in athletes' training programs and its potential to enhance athletes' performance under pressure.

In this study, we observed that drop height significantly affected the performance of American football athletes in executing DJs. Specifically, compared with the 30 cm drop height, the 40 cm and 50 cm heights resulted in lower jump heights and RFD, as well as lower RSI under both baseline and MF conditions, while higher values were observed for the K_{leg} indicator. These observations reveal the significant impact of drop height on athletes' neuromuscular coordination and energy conversion efficiency, consistent with our hypothesis 3. The study results are similar to those of *Bobbert, Huijing & van Ingen Schenau (1987)*, who emphasized the impact of drop height on muscle activation regulation and athletic performance by comparing the differences in DJ performance at drop heights of 20–60 cm. This further supports our conclusion and highlights the importance of considering drop height in training for athlete performance and health.

From a biomechanical perspective, increasing the drop height requires athletes to absorb more impact force during the landing phase, which may affect their subsequent takeoff ability. This phenomenon is consistent with the research of *Lesinski et al. (2018)*, who suggested that higher drop heights may prompt athletes to adopt more conservative movement strategies to reduce the risk of injury. Additionally, the higher K_{leg} indicates that athletes may increase muscle pre-activation to improve the absorption of impact force, which to some extent maintains their jumping performance. This is consistent with the research results of *Peng (2011)*, who pointed out that jumping from a height exceeding 40 cm offers no advantage in SSC power output and K_{leg} and suggested avoiding DJ from heights exceeding 60 cm, as it lacks biomechanical efficiency and may increase the risk of injury. This is also consistent with previous studies on the adaptive changes of the muscle–tendon complex in response to greater impact forces, which may involve changes in muscle fiber recruitment patterns, motor unit activation sequences, and intramuscular coordination and synchronization (*Dyhre-Poulsen, Simonsen & Voigt, 1991*; *Chen et al., 2022*).

Furthermore, the lower jump height and RFD, as well as the lower RSI, suggest that at higher drop heights, athletes may rely more on muscle strength rather than speed to complete the movement, which may be a natural response to reduce the impact force upon ground contact. This strategic change is consistent with previous studies observing muscle activity patterns, where athletes adjust their muscle activation patterns to adapt to different movement demands during high-intensity jumping actions (*Lesinski et al., 2018*). However, these findings are inconsistent with the previous research of *Matic et al. (2015)*,

who found that the optimal load height for athletes with high lower limb strength was 50 cm, and for those with low lower limb strength was 35 cm through experiments with 8 drop heights. In addition, *Di Giminiani & Petricola (2016)* found that the drop height for male athletes to achieve the maximum average power output in DJ was around 40 cm by comparing the power output at drop heights of 20 cm to 60 cm. These differences may stem from the characteristics of the subject groups; Matic et al. and Giminiani et al. studied male students rather than professional athletes, and not unified in a single sport, which may have affected the universality of the results. Determining the optimal drop height in DJ training is a complex issue influenced by various variables, including experimental design, data collection and analysis strategies, and athletes' lower limb strength capabilities (*Zang et al., 2019*). The data from this study suggest that for American football athletes, a 30 cm drop height may provide a more appropriate training load, as American football athletes often need to perform multidirectional jumps and change movements with rapid reaction characteristics in daily training and matches, rather than a single maximum force output. The explosive power and coordination required by the 30 cm drop height may be closer to the actual needs of American Football, making this height an effective training load to improve their competitive performance.

In summary, this study reveals that both MF and drop height are key factors affecting the lower limb movement performance of American football athletes when executing DJs. These findings are significant for developing targeted training programs, especially considering individual differences and training goals of athletes. Future research should further explore the impact of different drop heights on athletes' technical and physiological adaptations to better understand the mechanisms behind these phenomena and provide more personalized training recommendations for athletes. By gaining a deeper understanding of the impact of drop height on athletic performance, more scientific training guidance can be provided for athletes, optimizing training effects and reducing the risk of sports injuries.

This study has several limitations. First, although the sample size ($n = 12$) met the minimum requirement based on our a priori power calculation, it remained relatively small and consisted exclusively of male collegiate American football athletes. This may limit the generalizability of the findings to athletes of different genders, age groups, skill levels, and competitive backgrounds. Future research should therefore recruit larger and more diverse samples to enhance statistical power and external validity. Second, the conclusions were based on controlled laboratory conditions, with drop heights restricted to 30–50 cm, leaving uncertainty as to whether the findings can be extended to real competitive environments or to other drop heights. Third, mental fatigue was assessed primarily through a subjective workload scale, and the control condition employed a no-intervention design. Future studies could incorporate physiological or behavioral indicators of mental fatigue and utilize a low-cognitive-demand active control task to improve internal validity and the precision of interpretations regarding fatigue-related effects.

CONCLUSIONS

This study investigated the impact of MF and drop height on the DJ performance of American football athletes. Under conditions of MF, a decline in average power output,

RSI, and RSR was observed in athletes, suggesting reduced efficiency in central nervous system function. Additionally, increasing the drop height to 40 cm or above significantly reduced jump height and RFD, and increased K_{leg}, indicating that athletes may adopt more conservative movement strategies to reduce the risk of injury. However, no interactive effects between MF and drop height were observed. These findings emphasize the importance of considering the impact of MF and drop height on athlete performance in sports that demand high intensity and complex decision-making. Future training and research should focus on how to mitigate the negative effects of MF through strategic training interventions and optimize drop height to enhance athletic performance and reduce the risk of injury.

ADDITIONAL INFORMATION AND DECLARATIONS

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Competing Interests

The authors declare there are no competing interests.

Author Contributions

- Zilong Wang conceived and designed the experiments, performed the experiments, analyzed the data, prepared figures and/or tables, authored or reviewed drafts of the article, and approved the final draft.
- Ziqi Feng conceived and designed the experiments, analyzed the data, prepared figures and/or tables, authored or reviewed drafts of the article, and approved the final draft.
- Mengya Lu conceived and designed the experiments, performed the experiments, analyzed the data, prepared figures and/or tables, authored or reviewed drafts of the article, and approved the final draft.
- Jing Sun conceived and designed the experiments, authored or reviewed drafts of the article, and approved the final draft.
- Tao Liu conceived and designed the experiments, analyzed the data, authored or reviewed drafts of the article, and approved the final draft.
- Qiuxia Zhang conceived and designed the experiments, analyzed the data, authored or reviewed drafts of the article, and approved the final draft.
- Xiangdong Wang conceived and designed the experiments, analyzed the data, authored or reviewed drafts of the article, and approved the final draft.

Human Ethics

The following information was supplied relating to ethical approvals (i.e., approving body and any reference numbers):

The study was granted approval by the Ethics Committee of Soochow University (Ethics Number: SUDA20240626H03).

Data Availability

The following information was supplied regarding data availability:

All raw data from the experiments are available in the Supplemental File.

Supplemental Information

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REFERENCES

Abbott W, Brownlee TE, Naughton RJ, Clifford T, Page R, Harper LD. 2020. Changes in perceptions of mental fatigue during a season in professional under-23 English Premier League soccer players. *Research in Sports Medicine* **28**:529–539 DOI 10.1080/15438627.2020.1784176.

Badin OO, Smith MR, Conte D, Coutts AJ. 2016. Mental fatigue: impairment of technical performance in small-sided soccer games. *International Journal of Sports Physiology and Performance* **11**:1100–1105 DOI 10.1123/ijspp.2015-0710.

Bobbert MF, Huijing PA, Van Ingen Schenau GJ. 1987. Drop jumping, II. The influence of dropping height on the biomechanics of drop jumping. *Medicine and Science in Sports and Exercise* **19**:339–346 DOI 10.1249/00005768-198708000-00004.

Bosco C, Luhtanen P, Komi PV. 1983. A simple method for measurement of mechanical power in jumping. *European Journal of Applied Physiology and Occupational Physiology* **50**:273–282 DOI 10.1007/BF00422166.

Brown DMY, Graham JD, Innes KI, Harris S, Flemington A, Bray SR. 2020. Effects of prior cognitive exertion on physical performance: a systematic review and meta-analysis. *Sports Medicine* **50**:497–529 DOI 10.1007/s40279-019-01204-8.

Byrne PJ, Moran K, Rankin P, Kinsella S. 2010. A comparison of methods used to identify 'optimal' drop height for early phase adaptations in depth jump training. *Journal of Strength and Conditioning Research* **24**:2050–2055 DOI 10.1519/JSC.0b013e3181d8eb03.

Brysbaert M. 2019. How many participants do we have to include in properly powered experiments? A tutorial of power analysis with reference tables. *Journal of Cognition* **2**:16 DOI 10.5334/joc.72.

Chen X, Dong X, Feng Y, Jiao Y, Yu J, Song Y, Li X, Zhang L, Hou P, Xie P. 2022. Muscle activation patterns and muscle synergies reflect different modes of coordination during upper extremity movement. *Frontiers in Human Neuroscience* **16**:912440 DOI 10.3389/fnhum.2022.912440.

Davidow D, Smith M, Ross T, James L, Paul L, Lambert M, Jones B, Hendricks S. 2023. Mental fatigue impairs tackling technique in amateur rugby union players. *International Journal of Sports Physiology and Performance* **18**:960–967 DOI 10.1123/ijspp.2023-0159.

Decker MJ, Torry MR, Wyland DJ, Sterett WI, Steadman JR. 2003. Gender differences in lower extremity kinematics, kinetics and energy absorption during landing. *Clinical Biomechanics* **18**:662–669 DOI 10.1016/S0268-0033(03)00090-1.

Di Giminiani R, Petricola S. 2016. The power output–drop height relationship to determine the optimal dropping intensity and to monitor the training intervention. *Journal of Strength and Conditioning Research* **30**:117–125 DOI 10.1519/JSC.0000000000001076.

Díaz-García J, López-Gajardo MÁ, Parraca JA, Batalha N, López-Rodríguez R, Ring C. 2024. Brain endurance training improves and maintains chest press and squat jump performance when fatigued. *Journal of Strength and Conditioning Research* **38**:1568–1575 DOI 10.1519/JSC.0000000000004847.

Dyhre-Poulsen P, Simonsen EB, Voigt M. 1991. Dynamic control of muscle stiffness and H reflex modulation during hopping and jumping in man. *The Journal of Physiology* **437**:287–304 DOI 10.1113/jphysiol.1991.sp018596.

Faul F, Erdfelder E, Lang A-G, Buchner A. 2007. G*Power 3: a flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods* **39**:175–191 DOI 10.3758/BF03193146.

Filipas L, Ferioli D, Banfi G, La Torre A, Vitale JA. 2021. Single and combined effect of acute sleep restriction and mental fatigue on basketball free-throw performance. *International Journal of Sports Physiology and Performance* **16**:415–420 DOI 10.1123/ijspp.2020-0142.

Furuhashi Y, Hioki Y, Yoshimoto S, Hayashi R. 2021. Effect of neutral, internal, and external focus on drop jump performance: is drop jump performance affected by plyometric training experience? *Journal of Strength and Conditioning Research* **35**:2874–2880 DOI 10.1519/JSC.0000000000004147.

Gantois P, Caputo Ferreira ME, Lima-Junior D, Nakamura FY, Batista GR, Fonseca FS, Fortes LS. 2020. Effects of mental fatigue on passing decision-making performance in professional soccer athletes. *European Journal of Sport Science* **20**:534–543 DOI 10.1080/17461391.2019.1656781.

Griffin A, Kenny IC, Comyns TM, Purtill H, Tiernan C, O'Shaughnessy E, Lyons M. 2021. Training load monitoring in team sports: a practical approach to addressing missing data. *Journal of Sports Sciences* **39**:2161–2171 DOI 10.1080/02640414.2021.1923205.

Hart SG, Staveland LE. 1988. Development of NASA-TLX (Task Load Index): results of empirical and theoretical research. *Advances in Psychology* **52**:139–183 DOI 10.1016/S0166-4115(08)62386-9.

Healy R, Kenny IC, Harrison AJ. 2016. Assessing reactive strength measures in jumping and hopping using the Optojump™ system. *Journal of Human Kinetics* **54**:23–32.

Healy R, Kenny IC, Harrison AJ. 2018. Reactive strength index: a poor indicator of reactive strength? *International Journal of Sports Physiology and Performance* **13**:802–809 DOI 10.1123/ijspp.2017-0511.

Hoffman JR. 2008. The applied physiology of American football. *International Journal of Sports Physiology and Performance* **3**:387–392 DOI 10.1123/ijspp.3.3.387.

Horita T, Komi PV, Nicol C, Kyröläinen H. 2002. Interaction between pre-landing activities and stiffness regulation of the knee joint musculoskeletal system in the drop jump: implications to performance. *European Journal of Applied Physiology* **88**:76–84 DOI 10.1007/s00421-002-0673-6.

Lesinski M, Prieske O, Beurskens R, Behm D, Granacher U. 2018. Effects of drop-height and surface instability on jump performance and knee kinematics. *International Journal of Sports Medicine* **39**:50–57 DOI 10.1055/s-0043-117610.

Leukel C, Gollhofer A, Keller M, Taube W. 2008. Phase- and task-specific modulation of soleus H-reflexes during drop-jumps and landings. *Experimental Brain Research* **190**:71–79 DOI 10.1007/s00221-008-1450-5.

Lima-Junior D, Fortes LS, Ferreira MEC, Gantois P, Barbosa BT, Albuquerque MR, Fonseca FS. 2024. Effects of smartphone use before resistance exercise on inhibitory control, heart rate variability, and countermovement jump. *Applied Neuropsychology: Adult* **31**:48–55 DOI 10.1080/23279095.2021.1990927.

Liu H, Liu Z, Ha J, Jin Z, Di S. 2020. A study on the characteristics, influencing factors, and monitoring strategies of rugby injuries. *Journal of Wuhan Sports University* **5**:75–81 DOI 10.15930/j.cnki.wtxb.2020.05.011.

Lopes TR, Oliveira DM, Simurro PB, Akiba HT, Nakamura FY, Okano AH, Dias ÁM, Silva BM. 2020. No sex difference in mental fatigue effect on high-level runners' aerobic performance. *Medicine and Science in Sports and Exercise* **52**:2207–2216 DOI 10.1249/MSS.0000000000002346.

MacDonald LA, Minahan CL. 2016. Indices of cognitive function measured in rugby union players using a computer-based test battery. *Journal of Sports Sciences* **34**:1669–1674 DOI 10.1080/02640414.2015.1132003.

Mangin T, Audiffren M, Lorcery A, Mirabelli F, Benraiss A, André N. 2022. A plausible link between the time-on-task effect and the sequential task effect. *Frontiers in Psychology* **13**:998393 DOI 10.3389/fpsyg.2022.998393.

Marcora SM, Staiano W, Manning V. 2009. Mental fatigue impairs physical performance in humans. *Journal of Applied Physiology* **106**:857–864 DOI 10.1152/japplphysiol.91324.2008.

Mariano Y, Martin K, Mara J. 2023. Mental fatigue and technical performance in elite rugby league. *Journal of Sports Sciences* **41**:584–595 DOI 10.1080/02640414.2023.2228138.

Martin K, Staiano W, Menaspà P, Hennessey T, Marcora S, Keegan R, Thompson KG, Martin D, Halson S, Rattray B. 2016. Superior inhibitory control and resistance to mental fatigue in professional road cyclists. *PLOS ONE* **11**:e0159907 DOI 10.1371/journal.pone.0159907.

Matic MS, Pazin NR, Mrdakovic VD, Jankovic NN, Ilic DB, Stefanovic DL. 2015. Optimum drop height for maximizing power output in drop jump: the effect of maximal muscle strength. *Journal of Strength and Conditioning Research* **29**:3300–3310 DOI 10.1519/JSC.0000000000001018.

Molina R, Denadai BS. 2012. Dissociated time course recovery between rate of force development and peak torque after eccentric exercise. *Clinical Physiology and Functional Imaging* **32**:179–184 DOI 10.1111/j.1475-097X.2011.01074.x.

Moreira A, Aoki MS, Franchini E, Da Silva Machado DG, Paludo AC, Okano AH. 2018. Mental fatigue impairs technical performance and alters neuroendocrine and autonomic responses in elite young basketball players. *Physiology & Behavior* **196**:112–118 DOI 10.1016/j.physbeh.2018.08.015.

Mrdakovic V, Ilic DB, Jankovic N, Rajkovic Z, Stefanovic D. 2008. Pre-activity modulation of lower extremity muscles within different types and heights of deep jump. *Journal of Sports Science & Medicine* **7**:269–278.

Peng HT. 2011. Changes in biomechanical properties during drop jumps of incremental height. *Journal of Strength and Conditioning Research* **25**:2510–2518 DOI 10.1519/JSC.0b013e318201bcb3.

Queiros VS, Dantas M, Fortes LS, Silva LFD, Silva GMD, Dantas PMS, Cabral BGAT. 2021. Mental fatigue reduces training volume in resistance exercise: a cross-over and randomized study. *Perceptual and Motor Skills* **128**:409–423 DOI 10.1177/0031512520958935.

Ramirez-Campillo R, García-Hermoso A, Moran J, Chaabene H, Negra Y, Scanlan AT. 2022. The effects of plyometric jump training on physical fitness attributes in basketball players: a meta-analysis. *Journal of Sport and Health Science* **11**:656–670 DOI 10.1016/j.jshs.2020.12.005.

Robineau J, Lacome M, Piscione J, Bigard X, Babault N. 2017. Concurrent training in rugby sevens: effects of high-intensity interval exercises. *International Journal of Sports Physiology and Performance* **12**:336–344 DOI 10.1123/ijspp.2015-0370.

Russell S, Jenkins D, Halson S, Kelly V. 2020. Changes in subjective mental and physical fatigue during netball games in elite development athletes. *Journal of Science and Medicine in Sport* **23**:615–620 DOI 10.1016/j.jsams.2019.12.017.

Schampheleer E, Roelands B. 2024. Mental fatigue in sport—from impaired performance to increased injury risk. *International Journal of Sports Physiology and Performance* **19**:1–9 DOI 10.1123/ijspp.2023-0527.

Slater A, Campbell A, Smith A, Straker L. 2015. Greater lower limb flexion in gymnastic landings is associated with reduced landing force: a repeated measures study. *Sports Biomechanics* **14**:45–56 DOI 10.1080/14763141.2015.1029514.

Smith MR, Chai R, Nguyen HT, Marcora SM, Coutts AJ. 2019. Comparing the effects of three cognitive tasks on indicators of mental fatigue. *The Journal of Psychology* **153**:759–783 DOI 10.1080/00223980.2019.1611530.

Smith MR, Coutts AJ, Merlini M, Deprez D, Lenoir M, Marcora SM. 2016. Mental fatigue impairs soccer-specific physical and technical performance. *Medicine and Science in Sports and Exercise* **48**:267–276 DOI 10.1249/MSS.0000000000000762.

Smith MR, Thompson C, Marcora SM, Skorski S, Meyer T, Coutts AJ. 2018. Mental fatigue and soccer: current knowledge and future directions. *Sports Medicine* **48**:1525–1532 DOI 10.1007/s40279-018-0908-2.

Sotiropoulos K, Smilios I, Barzouka K, Christou M, Bogdanis G, Douda H, Tokmakidis SP. 2023. Effects of drop jump training from different heights and weight training on vertical jump and maximum strength performance in female volleyball players. *Journal of Strength and Conditioning Research* **37**:423–431 DOI 10.1519/JSC.0000000000004272.

Staiano W, Bonet LRS, Romagnoli M, Ring C. 2024. Mental fatigue impairs repeated sprint and jump performance in team sport athletes. *Journal of Science and Medicine in Sport* **27**:105–112 DOI 10.1016/j.jsams.2023.10.016.

Thompson CJ, Fransen J, Skorski S, Smith MR, Meyer T, Barrett S, Coutts AJ. 2019. Mental fatigue in football: is it time to shift the goalposts? An evaluation of the current methodology. *Sports Medicine* **49**:177–183 DOI 10.1007/s40279-018-1016-z.

Thompson CJ, Noon M, Towlson C, Perry J, Coutts AJ, Harper LD, Skorski S, Smith MR, Barrett S, Meyer T. 2020. Understanding the presence of mental fatigue in English academy soccer players. *Journal of Sports Sciences* **38**:1524–1530 DOI 10.1080/02640414.2020.1746597.

Wan X, Li Q, Yang C, Liu H, Qu F. 2017. Comparison of five different methods for calculating vertical jump height. *Journal of Beijing Sport University* **40**:62–66 DOI 10.19582/j.cnki.11-3785/g8.2017.11.011.

Wang IL, Chen YM, Zhang KK, Li YG, Su Y, Wu C, Ho CS. 2021. Influences of different drop height training on lower extremity kinematics and stiffness during repetitive drop jump. *Applied Bionics and Biomechanics* **2021**:5551199 DOI 10.1155/2021/5551199.

Wang Z, Meng L, Lu M, Kong L, Xue J, Zhang Z, Meng X, Zhang Q. 2024. Effects of attentional focus strategies in drop landing biomechanics of individuals with unilateral functional ankle instability. *Frontiers in Physiology* **15**:1444782 DOI 10.3389/fphys.2024.1444782.

Wang Z, Zhang Q, Lu M, Liu H, Xue J, Kong L, Zheng Y, Meng L, Zhang Z. 2025. The impact of combined physical and mental fatigue on lower limb explosive power and heart rate variability in rugby players. *China Sports Science and Technology* **61**:3–13 DOI 10.16470/j.csst.2024120.

Weerakkody NS, Taylor CJ, Bulmer CL, Hamilton DB, Gloury J, O'Brien NJ, Saunders JH, Harvey S, Patterson TA. 2021. The effect of mental fatigue on the performance of Australian football specific skills amongst amateur athletes. *Journal of Science and Medicine in Sport* **24**:592–596 DOI 10.1016/j.jsams.2020.12.003.

Yang C, Shi L, Lu Y, Wu H, Yu D. 2024. Post-activation performance enhancement of countermovement jump after drop jump *versus* squat jump exercises in elite rhythmic gymnasts. *Journal of Sports Science & Medicine* **23**:611–618 DOI 10.52082/jssm.2024.611.

You CH, Huang CH. 2022. Effects of leg stiffness regulated by different landing styles on vertical drop jump performance. *Journal of Human Kinetics* **83**:29–37 DOI 10.2478/hukin-2022-0066.

Zang Y, Xu Y, Yin F, Zhou H, Feng X. 2019. The influence of landing height on the lower limb biomechanics and reactive strength of elite sprinters during drop jumps. *Physical Education and Science* **6**:101–110 DOI 10.13598/j.issn1004-4590.2019.06.014.